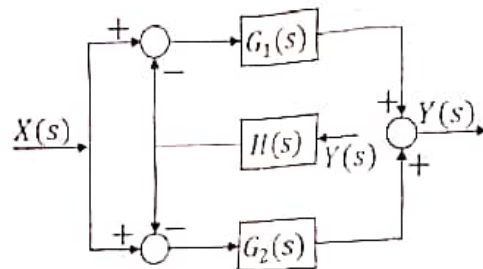
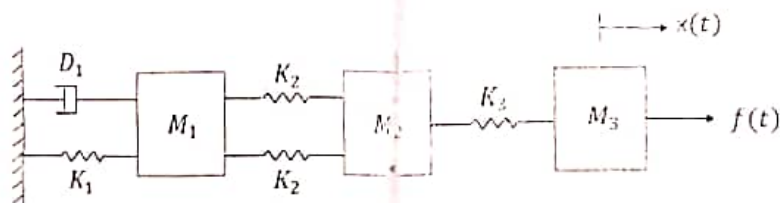


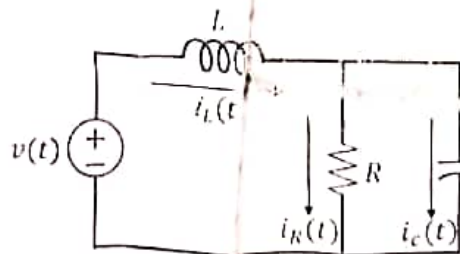
1. Considering the system interconnection in the below figure, find the overall transfer function from input $x(t)$ to the output $y(t)$. [6]



2. For the mechanical system shown below, write down the equations in 's'-domain to completely describe the dynamics of the system. Draw an electric analog of this mechanical system. [6 + 2]



3. Given the electrical network in the below figure, find a state-space representation of the system dynamics considering $v(t)$ as the input and $i_L(t)$ as the output. [6]



Handwritten solution for the state-space representation:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}v(t)$$

$$\mathbf{A} = \begin{bmatrix} -\frac{R}{L} & \frac{1}{L} \\ \frac{1}{C} & -\frac{1}{RC} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix}$$

where $\mathbf{x} = \begin{bmatrix} i_L \\ i_C \end{bmatrix}$.

(a) Sketch the Nyquist contour (in s -plane) and the corresponding Nyquist plot for the loop transfer function

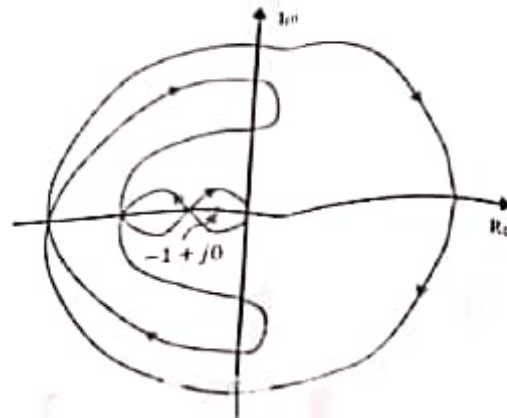
$$L(s) = G(s)H(s) = \frac{K(s+3)}{s(s-1)}, \quad K > 1$$

and apply Nyquist stability criterion to conclude on the closed-loop stability.

(b) Considering $K = 2$, find the gain margin for the system considered in Part (a). What do you interpret from the obtained gain margin value in terms of the effect of increasing gain (K) on the closed-loop stability?

(c) Below is the Nyquist plot for a system having 1 (one) right half plane open-loop pole. Check the stability of the closed-loop with unity negative feedback from the plot.

[8 + 4 + 2]



6. (a) Draw a rough asymptotic Bode magnitude plot (phase plot not required) for the transfer function

$$G(s) = \frac{25(s+2)(s+6)}{s(s+1)(s+5)(s+10)}$$

Clearly mention the corner frequencies, corresponding gains at corner frequencies, and slopes of each part.

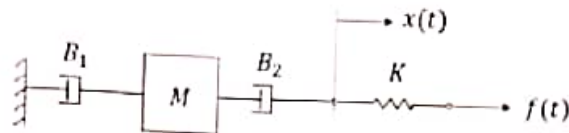
(b) The loop transfer function of a feedback control system is

$$L(s) = \frac{K}{s(s^2 + 4s + 5)}$$

Sketch the root locus with respect to K for the equation $1 + L(s) = 0$. Provide all the required details including the centroid, angle of asymptotes, break-away/break-in points, angle of departure/arrival (wherever applicable).

[6 + 6]

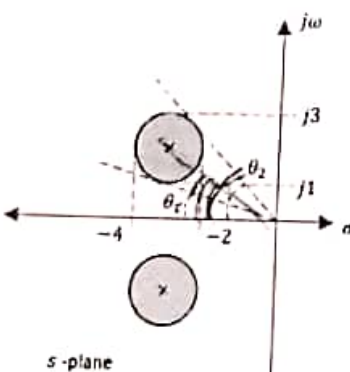
1. Consider the mechanical system shown below with the parameter values $M = 10$ kg, $K = 2$ N-m, $B_1 = 2$ N-s/m, and $B_2 = 5$ N-s/m.



- (a) Find the transfer function $\frac{X(s)}{F(s)}$.
 (b) Is the transfer function BIBO stable?

[6 + 1]

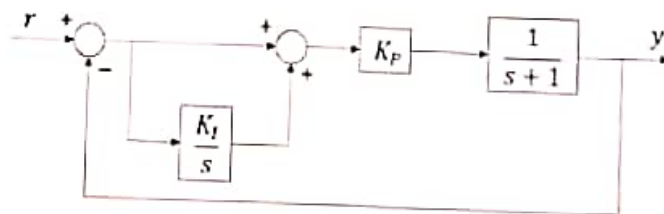
2. (a) For a second order system in standard form, give an estimate of the range of values of *natural frequency* and *damping ratio* that would correspond to the shaded circular regions in s-plane as shown in the below figure (a).



(a)

- (b) Find values of K_P and K_I for the unity feedback system depicted in the below figure (b) such that the closed-loop poles lie within the shaded region of the s-plane, preferably at the centers.

[4 + 4]



(b)

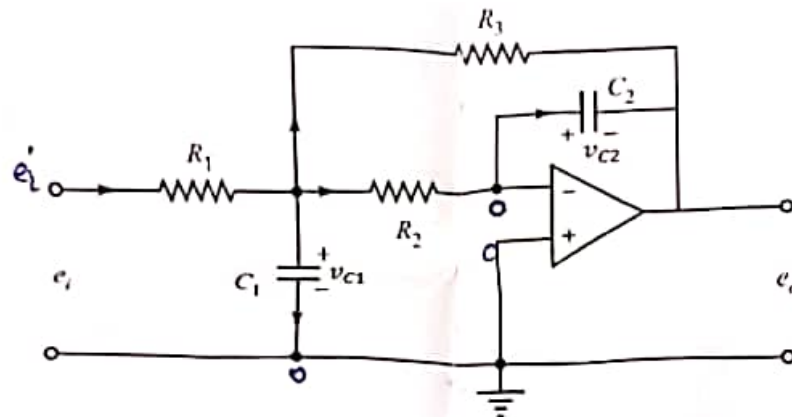
3. The open-loop transfer function of a unity gain negative feedback system is

$$G(s) = \frac{s^3 + 7s^2 - 21s + 10}{s^6 + s^5 - 6s^4 - s^3 - 8s^2 + 20s - 4}$$

- (a) Find the number of closed-loop poles in the open left half and open right half of s-plane.
 (b) If there are purely imaginary closed-loop poles, find their locations.

[5 + 2]

4. (a) Considering $v_{C1}(t)$ and $v_{C2}(t)$ as state variables, obtain the state-space representation of the given operational-amplifier circuit with $e_i(t)$ as the input and $e_o(t)$ as the output.



- (b) By obtaining the state transition matrix, find the expression of $e_o(t)$, for all $t \geq 0$ when $e_i(t) = 0 \quad \forall t \geq 0$, $v_{C1}(0) = 2 \text{ V}$, and $v_{C2}(0) = -1 \text{ V}$. Take $R_1 = R_3 = 1 \Omega$, $R_2 = 0.5 \Omega$, $C_1 = 1 \text{ F}$, and $C_2 = 2 \text{ F}$. [5 + 5]

5. (a) Linearize the following differential equation for small excursions of x near π :

$$\frac{d^2x}{dt^2} + 3 \frac{dx}{dt} + (2x - \sin x) = 0.$$

- (b) For a given initially relaxed LTI system, the output to an unit step input is represented by.

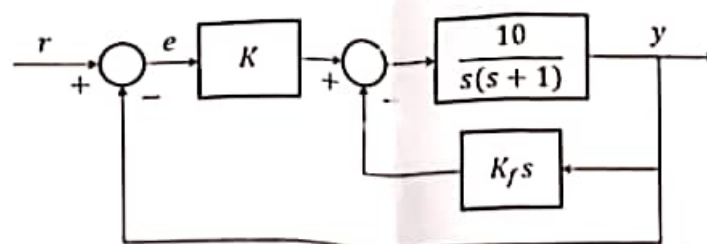
$$y(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \cos(\omega_n \sqrt{1-\zeta^2} t - \phi), \quad t \geq 0.$$

where $\phi = \tan^{-1} \left(\frac{\zeta}{\sqrt{1-\zeta^2}} \right)$. Find the impulse response $h(t)$ of the system, and draw the nature of response when $\zeta\omega_n > 0$, $0 < \zeta < 1$. [3 + 3]

6. Consider the following feedback control system configuration where two controller parameters K and K_f are to be designed to meet the closed-loop performance specifications:

- (i) damping ratio (ζ) = 0.5; (ii) velocity error constant (K_v) = 10.

Design K and K_f . [6]



the key Consider a unity negative feedback system with forward path transfer function $G(s) = \frac{1}{(s+4)^3}$.

(a) Find the location of the dominant closed-loop poles to yield a 1.6 second settling time and an overshoot of 25%.

(b) If a compensator with a zero at -1 is used to achieve the conditions of Part (a), what must be the angular contribution of the compensator pole at the second quadrant dominant pole location?

(c) Find the location of the compensator pole.

(d) Find the compensator gain required to meet the specifications stated in Part (a).

(e) Find the location of other closed-loop poles for the compensated system.

[2 + 5]

7. (a) Analyze controllability and observability of the system realization:

$$\dot{x}(t) = \begin{bmatrix} -7 & -10 \\ 5 & 8 \end{bmatrix} x(t) + \begin{bmatrix} -2 \\ 2 \end{bmatrix} u(t), \quad x_0 := x(0),$$

$$y(t) = \begin{bmatrix} 1 & 1 \end{bmatrix} x(t).$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

(b) Let $\{A, B, C, D\}$ be an n -th order realization of a given SISO transfer function $G(s) = \frac{b(s)}{a(s)}$, where $a(s) = s^n + a_1s^{n-1} + a_2s^{n-2} + \dots + a_n$. Find the matrices A and B when the controllability matrix $C(A, B) = I_n$, where I_n is the identity matrix of dimension $n \times n$.

[4 + 6]

$$\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

8. For the following system, the observer-based state-feedback controller design methodology has been adopted.

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

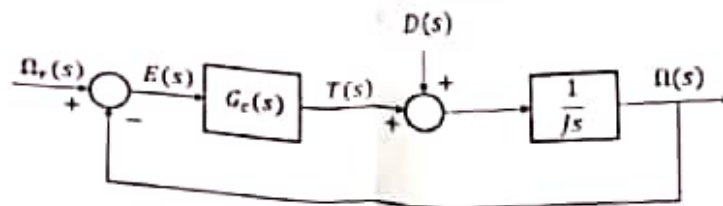
$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} x.$$

(a) Design the observer and controller gain matrices (L and K , respectively) to place the observer poles at $s = -5, -6$, and the dominant closed-loop poles at $s = -2, -3$.

(b) Derive the simplified transfer function of the overall closed-loop system.

[6 + 4]

1. The block-diagram below shows a car's speed control system which is subjected to a torque disturbance $d(t)$. In the block-diagram, $\Omega_r(s)$, $\Omega(s)$, $T(s)$, and $D(s)$ are the Laplace transforms of the reference speed, output speed, driving torque, and disturbance torque, respectively.



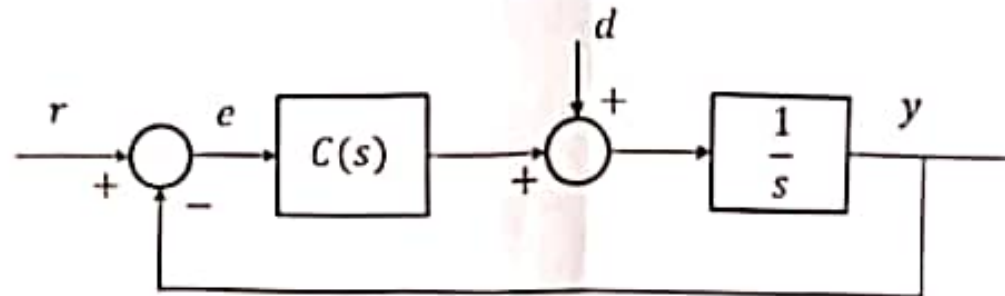
Consider that a PI controller, $G_c(s) = K_P + \frac{K_I}{s}$, is designed for the car to track the reference speed.

- Find the transfer function from the torque disturbance $D(s)$ to the speed error $E(s)$.
 - In the absence of the reference speed, verify whether the speed error $e(t)$ is eliminated at steady-state due to any step change in the torque disturbance. [4 + 4]
2. A unity negative feedback control system has an open-loop transfer function $G(s) = \frac{2(s+z)}{s(s+2)}$, where $z > 0$.
- Find the closed-loop transfer function and its settling time (t_s).
 - Calculate the sensitivity of the steady-state error e_{ss} for unit ramp input to changes in the parameter z .
 - With the help of root-locus, justify whether or not the system can be stabilized by a proportional controller having negative gain value. [3 + 3 + 2]
3. The loop transfer function of a negative feedback control system is given by

$$L(s) = G(s)H(s) = \frac{K(1+s)}{s^2(1+0.316s)}$$

- Determine analytically the possible maximum phase margin and the frequency at which it occurs.
- What should be the corresponding gain K for this phase margin? [4 + 4]

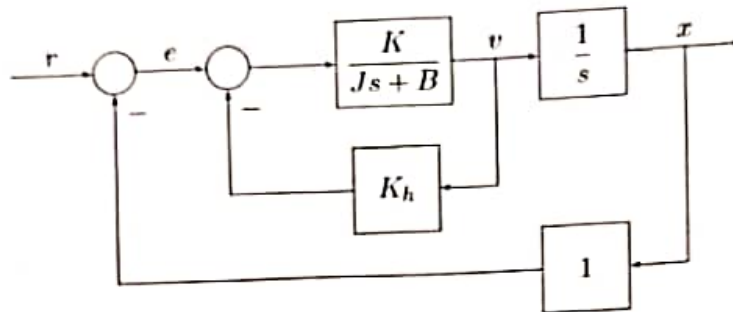
7. Answer the questions below for the following unity feedback system where the controller $C(s) = \frac{K}{s^2 + as + b}$ with parameters $K, a, b \in \mathbb{R}$.



- (a) Find the steady-state error due to a unit ramp reference input $r(t)$ in the absence of disturbance signal $d(t)$. Comment on the choice of the controller parameters to make the steady-state error zero.
- (b) Can you design the controller parameters of $C(s)$ which would result zero steady-state output ($y(\infty) = 0$) due to a unit step disturbance (keeping $r(t) \equiv 0$)?

[4 + 2]

1. For the following feedback system, design the values of K and K_h such that the unit step response of the overall closed-loop system would have peak-time of 1 second and maximum overshoot of 20%. With these designed values, determine the rise-time and settling time. Consider $J = 1 \text{ kg-m}^2$, and $B = 1 \text{ N-m-s/rad}$. [6]



2. Consider the following state space representation of a system:

$$\dot{x} = \begin{bmatrix} 1 & -2 \\ 1 & -1 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u, \quad x_0 := x(0) \in \mathbb{R}^2,$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} x + u.$$

- (a) Find the impulse response of the system.
(b) Is the impulse response absolutely integrable?

[6]

3. Consider the following ODE that describes an LTI system with input $u(t)$ and output $y(t)$:

$$\ddot{y}(t) + 6\dot{y}(t) + 11y(t) = 4u(t).$$

Represent the system dynamics in state space equations

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$y(t) = Cx(t) + Du(t),$$

such that A is a diagonal matrix.

[8]